

EDUCATION	Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua University Beijing, China 2024.08 - Present B.E. in Computer Science and Technology • Overall GPA: 3.8/4.0. • TOEFL iBT (2026 Band Scale): 5.0/6.0 (Equivalent to 100/120, CEFR C1) • Selected Coursework: Machine Learning (A), Natural Language Processing (A), AI Principles & Techniques (A), Algorithm Design (A+), Operating Systems & Distributed Systems (A), Mathematics for CS & AI (A).
PUBLICATIONS	1. Huigen Ye, Hua Xu, An Yan, Yaoyang Cheng. Large Language Model-driven Large Neighborhood Search for Large-Scale MILP Problems. In <i>Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)</i>, Spotlight. 2. Huigen Ye, Hua Xu, An Yan. LLM-driven Streamlining Optimizer for Large-scale Mixed Integer Linear Programming Problems. <i>Submitted to Nature Communications</i> (Under Review).
RESEARCH EXPERIENCE	Research on Applications of LLMs in Mixed Integer Linear Programming and Combinatorial Optimization Research Assistant, Supervised by Prof. Hua Xu, Tsinghua University 2024.06 - Present • Scale neighborhood selection for MILP by evolving heuristic code via a dual-layer LLM agent, enabling zero-shot transfer from small training samples to ultra-large problems with 10^6 variables, generalizing to diverse combinatorial problems. • Explored RAG-augmented LLMs for tool-using and long-horizon decision-making in complex optimization pipelines, optimizing branch-and-bound via RAG-initialized code synthesis and iterative Bayesian refinement. • (Ongoing) Unify MILP encoding with LLM token spaces through learnable graph tokenizers, leveraging SFT for priority score prediction and GRPO RL with greedy construction to directly optimize solution quality.
SELECTED AWARDS	• Scholarship for Outstanding Scientific and Technological Innovation, Tsinghua University 2025.10 • Champion, Huawei Algorithm Competition (University Track) 2025.09 • Gold Medal, 2025 Beijing Collegiate Programming Contest 2025.04 • Gold Medal, 10th China Collegiate Programming Contest (CCPC), Jinan Site 2024.11 • Gold Medal, 40th National Olympiad in Informatics (NOI), China 2023.07
SKILLS	Languages: C++, Python (Proficient); Go, Scala, Bash (Familiar). Mathematics: Probability Theory, Stochastic Processes, Abstract Algebra, Combinatorics. Tools: PyTorch, LaTeX, Git, Docker.